AssetVal Pro v2

Model Card

Tangible Asset Valuation & Replace-or-Retain Decision Support System

AssetVal Pro v2 is a structured, transparent, and standards-compliant decision-support system for evaluating physical infrastructure and industrial assets. It integrates Life Cycle Cost analysis, multi-criteria decision modeling, and ISO 31000 risk management into a clear 7-step workflow — producing auditable, evidence-based recommendations.

Model Name	AssetVal Pro v2	**Author**	Ahmed Raslan
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Type	Decision Support System (DSS)	**Phone**	+971 52 289 9595
Domain	Infrastructure & Industrial Asset Management	**Language**	Bilingual — Arabic (RTL) / English (LTR)
		License	Proprietary — All rights reserved
Deployment	Single-file HTML/JS — no server required		

ISO 55000	API 570	ASTM E917	ISO 31000	FHWA LCCA	IEC 61511

Document Date: May 11, 2026

Table of Contents

1 — Executive Summary

AssetVal Pro v2 guides practitioners through a structured 7-step asset evaluation process. It combines internationally recognized financial analysis (NPV, EUAC, IRR) with risk assessment (ISO 31000) and multi-criteria scoring (MCDM) to produce a clear, auditable recommendation: **Retain, Upgrade, or Replace**. The model is fully transparent — every formula is documented, every weight is user-adjustable, and the final decision always rests with the human expert.

PROBLEM	SOLUTION	OUTCOME
• Asset degradation over time	• Structured 7-step evaluation wizard	• Evidence-based recommendation
• Complex retention vs. replacement decisions	• Three financial alternatives compared	• NPV / EUAC / IRR / Payback computed
• Multiple stakeholders, currencies & standards	• Risk + condition + compliance scoring	• Sensitivity analysis at ±4% discount
• Need for auditable, defensible decisions	• Bilingual, multi-currency, print-ready	• Full audit trail for management

2 — Model Overview & Purpose

The primary purpose of AssetVal Pro v2 is to reduce subjectivity in asset management decisions by providing a repeatable, documented, and standards-aligned evaluation framework. It is not an autonomous decision-making system — it is a **structured decision-support tool** that enhances the quality and traceability of expert judgment.

Core Design Principles

Transparency	All calculations visible & documented. No hidden scores or unexplained outputs.
Interoperability	7 currencies, bilingual Arabic/English, 10 asset types across 2 major paths.
Standards Alignment	Built on ISO 55000, API 570, ASTM E917, ISO 31000, and FHWA LCCA.
Human-in-the-Loop	The model produces a recommendation — the expert makes the final call.
Auditability	Every input, weight, and output is printable and traceable for management reporting.
Accessibility	Zero installation — single HTML file, deployable on any web server or locally.

3 — Intended Use & Target Users

Field Engineer Basic Mode	Needs a fast, clear go/no-go decision in the field. Uses pre-filled defaults and gets a color-coded recommendation without financial complexity.
Asset Manager Basic + Expert	Produces management-ready reports with NPV comparisons, risk matrices, and sensitivity analysis for board-level decisions and capital planning.
Technical Consultant Expert Mode	Full control over discount rate, inflation, MCDM weights, and sector presets. Produces detailed auditable outputs aligned with international standards.

Out-of-Scope Use Cases

- ✗ Financial securities or investment portfolio valuation
- ✗ Real estate or intangible asset valuation
- ✗ Autonomous asset replacement without human review
- ✗ Legal or regulatory compliance certification (standalone)
- ✗ Safety system analysis as a replacement for dedicated safety studies

4 — System Architecture — 7-Step Workflow

AssetVal Pro v2 follows a linear 7-step wizard. Each step gates the next, ensuring data completeness before calculations are performed. All processing is client-side — no data leaves the user's browser.

Step 1 Path & Type Selection	User selects Infrastructure (5 sub-types: Pipelines, Water, Power, Roads, Telecom) or Equipment (5 sub-types: Rotating, Static, Electrical, Instrumentation, Vehicles). Dynamic fields load per type; compliance badges update automatically.
Step 2 Basic Asset Data	Asset identity, installation year, design life, condition score (1-10), book value, and compliance status. Actual age and remaining life are auto-calculated.
Step 3 LCC Cost Inputs	Annual maintenance and operating costs, escalation rate, salvage value. Expert Mode unlocks discount rate (WACC) and inflation rate inputs.
Step 4 Benefit & Risk	Annual revenue/savings input and benefit growth (Expert Mode). Interactive 5×5 Risk Matrix with PoF/CoF selectors; Expert Mode adds safety and environmental risk.
Step 5 Three Alternatives	Parameterize Retain (A): improved maintenance + life extension; Upgrade (B): upgrade cost + new life + reduced maintenance; Replace (C): new asset cost + installation + new life + maintenance.
Step 6 MCDM Weights	Five criteria sliders (Financial, Condition, Risk, Obsolescence, Compliance) must sum to 100%. Four sector presets available: Oil & Gas, Utilities, Transport, Telecom.
Step 7 Results & Recommendation	Full financial comparison table, winner banner with MCDM score, cumulative cost chart (Chart.js, 3 alternatives), and sensitivity analysis table at 5 discount rate scenarios.

5 — Methodology & Algorithms

Four analytical methodologies are combined into a unified scoring pipeline. Each is independently interpretable and follows published international standards.

Life Cycle Cost Analysis (LCCA)

Evaluates total cost of ownership over an asset's life: capital, maintenance (with compound escalation), operating (with inflation), and salvage recovery. Basis: ASTM E917 & FHWA LCCA.

Net Present Value (NPV) Analysis

Discounts all future cash flows to present value using user-specified WACC. Enables fair comparison of alternatives with different lifespans via EUAC normalization.

Risk-Based Assessment (ISO 31000)

Quantifies failure risk via a 5×5 matrix (PoF × CoF). Produces a Risk Score (1-25) classified into four levels (Low / Medium / High / Critical) feeding the MCDM module.

Multi-Criteria Decision Model (MCDM)

Combines Financial, Condition, Risk, Obsolescence, and Compliance scores into a weighted composite (0-100) per alternative. Basis: ISO 55000 asset management principles.

6 — Financial Engine — Formulas & Calculations

Net Present Value (NPV)

```
NPV = SUM[(C_maint*(1+e)^y + C_op*(1+inf)^y)/(1+r)^y, y=1..n] - S/(1+r)^n
```

C_maint	Annual maintenance cost (base year)
C_op	Annual operating cost (base year)
e	Maintenance escalation rate (%/yr)
inf	Inflation rate (%/yr) — applied to operating cost
r	Discount rate / WACC
n	Asset life (years)
S	Salvage value — deducted at end of life

Equivalent Uniform Annual Cost (EUAC)

```
EUAC = NPV * r*(1+r)^n / ((1+r)^n - 1)
```

NPV	Net Present Value from above
r	Discount rate
n	Life of the alternative being evaluated

Internal Rate of Return (IRR)

```
Bisection method (60 iterations): SUM[Savings_y/(1+IRR)^y] - Cost_initial = 0
```

Cost_initial	Replacement cost + installation
Savings_y	Annual cost savings vs. current state
Bounds	Search range: -99% to +500%

Simple Payback Period

```
Payback = Cost_replacement / (Current_annual_cost - New_annual_cost)
```

Cost_replacement	New asset + installation
Current_annual	Current maintenance + operating
New_annual	New asset maintenance + operating

Default Assumptions — Basic Mode

Parameter	Default	Rationale
Discount Rate (r)	8%	Typical Gulf infrastructure WACC
Inflation Rate	3%	Regional CPI estimate
Maintenance Escalation	5%/yr	Industry norm for aging assets
IRR Search Bounds	-99% to +500%	Covers all practical scenarios
Salvage Value	User input	Asset-specific — no assumed default

7 — Risk Assessment Framework (ISO 31000)

Risk is quantified using the ISO 31000:2018 framework. The 5×5 matrix produces a Risk Score (PoF × CoF) from 1 to 25, classified into four action levels. This score feeds directly into the MCDM Risk criterion.

```
Risk Score = PoF × CoF
```

Score	Level	Action Required

1 – 4	**Low**	Monitor — normal maintenance cycle
5 – 9	**Medium**	Review — enhanced inspection recommended
10 – 16	**High**	Intervene — prioritize remediation planning
17 – 25	**Critical**	Immediate action — safety/operational risk

Probability of Failure (PoF) Scale

Level	Label	Description
1	Rare	< 1% probability over asset life
2	Unlikely	1-10% — unlikely under normal conditions
3	Possible	10-50% — could occur given current condition
4	Likely	50-90% — expected within planning horizon
5	Almost Certain	> 90% — failure imminent or overdue

8 — Multi-Criteria Decision Model (MCDM)

The MCDM module scores each alternative across five criteria and computes a weighted composite score (0-100). The alternative with the highest composite score receives the recommendation. All weights are user-adjustable and must sum to 100%.

Criterion	Default Weight	Min / Max	Scoring Basis
Financial (LCC)	30%	5% / 60%	Inverse normalized EUAC across alternatives
Technical Condition	25%	5% / 60%	Condition score (1-10) mapped to 0-100
Risk	25%	5% / 60%	Inverse normalized Risk Score (1-25)
Obsolescence	10%	0% / 30%	Age / Design Life ratio (Expert Mode)
Compliance	10%	0% / 30%	Full=100, Partial=50, Non-compliant=0

Sector Presets

Sector	Financial	Condition	Risk	Obsolescence	Compliance
Default	30%	25%	25%	10%	10%
Oil & Gas	25%	20%	35%	10%	10%
Utilities	30%	25%	25%	10%	10%
Transport	35%	30%	20%	10%	5%
Telecom	20%	20%	15%	35%	10%

MCDM Formula

```
Score = (w_fin*F + w_cond*C + w_risk*R + w_obs*O + w_comp*P) / SUM(weights)
```

Result range: 0 – 100. Higher score = more favorable alternative.

9 — Input / Output Specification

Key Inputs

Category	Field	Type	Required?
Identity	Asset ID, Name	Text	Yes
Lifecycle	Install Year, Design Life	Integer	Yes
Financial	Original Cost, Book Value	Decimal	Yes
LCC	Maint. Cost, Operating Cost, Escalation, Salvage	Decimal	Yes
Expert LCC	Discount Rate (WACC), Inflation Rate	Decimal	Expert Mode
Risk	PoF (1-5), CoF (1-5)	Integer	Yes
Alternatives	A/B/C Costs, Lifespans, Maintenance	Decimal	Yes
MCDM	5 Criteria Weights (sum=100%)	Integer	Yes
Type-Specific	Corrosion, PCI, MTBF, Load%, etc.	Mixed	By asset type

Outputs

- NPV for each alternative (in selected currency)
- EUAC — enables fair cross-lifespan comparison
- IRR — for replacement alternative (bisection method)
- Payback Period — years to recover replacement investment
- Risk Score (1-25) with level classification
- MCDM Composite Score (0-100) per alternative

- Winner recommendation with score justification
- Sensitivity table at ±2% and ±4% discount rate
- Cumulative cost chart (3 alternatives over project life)
- Print-ready report in Arabic or English

10 — Smart Validation & Warning Thresholds

Asset / Field	Threshold	Warning
Pipeline — Wall Thickness	< 80%	Immediate API 570 inspection required
Water Network — Leakage Rate	> 20%	High NRW — immediate intervention
Power Network — Load Ratio	> 90%	Overload — accelerates deterioration
Roads — PCI Index	< 40	Immediate resurfacing required
Equipment — Efficiency	< 70%	Urgent efficiency review required
All — Maintenance Escalation	> 20%/yr	Escalation too high — verify input
All — Annual Maint. Cost	> 50% of original	Strong replacement indicator
All — Condition Score	>= 8 / 10	Critical condition — urgent review

■ 11 — Model Limitations & Assumptions

■ **Data Quality**	The model is only as accurate as the inputs provided. Inaccurate maintenance cost history or unrealistic escalation rates will produce misleading outputs.
■ **Deterministic Estimates**	All financial inputs are point estimates. No Monte Carlo or probabilistic simulation is performed beyond the ±4% discount rate sensitivity table.
■ **No Real-Time Data**	Manual data entry only. No connection to CMMS, ERP, or IoT systems.
■ **Linear Life Estimation**	Remaining life = Design Life – Actual Age. Does not reflect condition-based life extension; expert override is expected.
■ **Currency Handling**	Multi-currency support is for labeling only. No FX conversion is performed. All inputs must be in the same currency.
■ **Subjective Inputs**	PoF, CoF, and MCDM weights require informed expert judgment. Range checks are applied but reasonableness is not algorithmically validated.

■ 12 — Transparency & Explainability

AssetVal Pro v2 is designed to be a fully explainable decision-support system, aligned with EU AI Act principles and ISO/IEC 42001 AI management standards.

✓ **Visible Formulas**	Every formula (NPV, EUAC, IRR, MCDM) is documented in contextual popovers within the tool and in this model card. Nothing is hidden.
✓ **Adjustable Weights**	All MCDM weights are user-visible and user-adjustable. The 100% constraint is enforced transparently on-screen.
✓ **Sensitivity Output**	Results at ±2% and ±4% discount rate show how stable the recommendation is.
✓ **Open Source Logic**	All logic is client-side JavaScript — inspectable by any technical reviewer.
✓ **Audit Trail**	Print report captures all inputs, weights, and outputs for regulatory audit.

13 — Confidence, Uncertainty & Human Oversight

AssetVal Pro v2 produces a **recommendation**, not a decision. The model aggregates quantitative data and expert weights into a ranked output, but the final asset management decision must always be reviewed and approved by a qualified human professional.

Model Output	Ranked recommendation (highest MCDM composite score)
Decision Authority	Asset manager / engineering team — not the model
Confidence Level	High for financial comparisons; moderate for MCDM (subjective weights)
Uncertainty Sources	Input data quality, PoF/CoF subjectivity, discount rate selection
Recommended Review	Cross-check IRR and payback against organizational investment thresholds
Override Protocol	Users may always override the recommendation and document their rationale

14 — Ethical Considerations

Non-Discrimination	Identical analytical logic applied to all asset types and users. No demographic, regional, or organizational bias is embedded in the model.
Environmental Impact	Optional environmental risk scoring in Expert Mode. Replacement decisions should consider lifecycle environmental costs where applicable.
Safety Priority	High safety risk scores should always trigger additional engineering review, regardless of the MCDM recommendation.
Data Privacy	All data processed locally in the browser. No data is transmitted, stored in the cloud, or shared with any third party.
Scope Integrity	Intended for professional engineering use. Misuse for financial reporting, securities valuation, or regulatory certification is explicitly out of scope.

15 — Compliance & Standards Reference

Standard	Full Title	Application in Model
ISO 55000:2014	Asset Management — Overview, Principles & Terminology	Core framework & MCDM criteria structure
API 570:2016	Piping Inspection Code	Pipeline fields & wall thickness warnings
ASTM E917	Standard Practice for Measuring LCC of Buildings	LCC methodology & formula structure
ISO 31000:2018	Risk Management — Guidelines	5×5 Risk Matrix, PoF/CoF definitions
FHWA LCCA	Life-Cycle Cost Analysis in Pavement Design	Road & bridge LCC methodology
IEC 61511	Functional Safety — SIS for Process Industry	Instrumentation SIL level field
ISO 42001:2023	AI Management Systems	Transparency & human oversight principles
EU AI Act (2024)	Regulation on Artificial Intelligence	Accountability & explainability alignment

16 — Author & Contact

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TOOL	AssetVal Pro v2 — Tangible Asset Valuation System
DATE	May 2026
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This model card was prepared in accordance with global AI documentation best practices including Google Model Cards (Mitchell et al., 2019), Hugging Face model card guidelines, and ISO/IEC 42001:2023. It serves as the official technical and transparency documentation for AssetVal Pro v2.